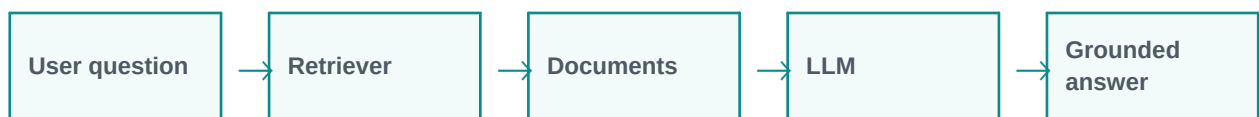


Retrieval Augmented Generation (RAG)

RAG combines search with generation.

Simple explanation: RAG means connecting an LLM with external documents or databases. Instead of answering only from memory, the system first retrieves relevant information and then generates a grounded answer.

Learning style: this handout starts with a very simple explanation and gradually increases the depth. It is designed for classroom training, participant reading, and quick revision.



What you will learn

- Meaning in simple words
- Key components and architecture
- Practical examples and use cases
- Where it fits in GenAI and agentic AI systems
- Benefits, challenges, and implementation points



1. Beginner-Friendly Explanation

RAG means connecting an LLM with external documents or databases. Instead of answering only from memory, the system first retrieves relevant information and then generates a grounded answer.

Think of it like this:

RAG is like asking a librarian to first find the right books, and then asking a writer to prepare the answer using those books.

Simple real-time examples

- Ask questions on HR policy
- Summarize customer tickets
- Search legal clauses
- Find product documentation
- Generate cited answers

2. Gradually Deeper Explanation

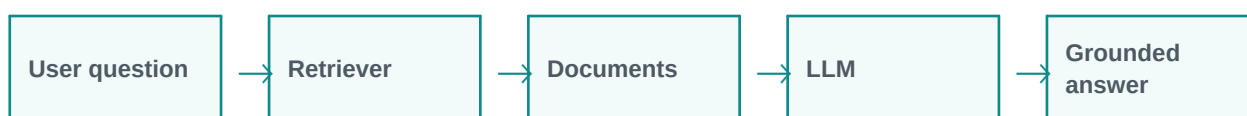
Retrieval Augmented Generation (RAG) becomes more powerful when we understand its components, data flow, limitations, and business fit. In real projects, the concept is rarely used alone. It usually works with prompts, models, tools, documents, APIs, evaluation, monitoring, and human review.

Important concepts

Concept	Meaning
PDF Q&A;	PDF Q&A; is an important part of understanding and applying Retrieval Augmented Generation (RAG) in practical GenAI systems.
Policy assistant	Policy assistant is an important part of understanding and applying Retrieval Augmented Generation (RAG) in practical GenAI systems.
Knowledge base search	Knowledge base search is an important part of understanding and applying Retrieval Augmented Generation (RAG) in practical GenAI systems.
Support case analysis	Support case analysis is an important part of understanding and applying Retrieval Augmented Generation (RAG) in practical GenAI systems.
Research assistant	Research assistant is an important part of understanding and applying Retrieval Augmented Generation (RAG) in practical GenAI systems.

3. Architecture / Flow

A typical architecture can be understood as a simple flow. The exact design changes based on the project, but the pattern below is a useful starting point.



Architecture explanation

Step 1: User question - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 2: Retriever - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 3: Documents - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 4: LLM - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 5: Grounded answer - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Common implementation layers

- User interface or API layer
- Prompt and instruction layer
- Model layer
- Tool/retrieval/data layer
- Validation, safety, and logging layer
- Human review or feedback layer

4. Detailed Examples

Below are example scenarios showing how this topic appears in real GenAI projects.

Scenario	How it works	Expected output
Ask questions on HR policy	The system receives the request, applies Retrieval Augmented Generation (RAG), and processes the task step by step.	A useful, structured, reviewable result.
Summarize customer tickets	The system receives the request, applies Retrieval Augmented Generation (RAG), and processes the task step by step.	A useful, structured, reviewable result.
Search legal clauses	The system receives the request, applies Retrieval Augmented Generation (RAG), and processes the task step by step.	A useful, structured, reviewable result.



Scenario	How it works	Expected output
Find product documentation	The system receives the request, applies Retrieval Augmented Generation (RAG), and processes the task step by step.	A useful, structured, reviewable result.
Generate cited answers	The system receives the request, applies Retrieval Augmented Generation (RAG), and processes the task step by step.	A useful, structured, reviewable result.



5. Benefits and Challenges

Benefits	Challenges / Risks
Improves productivity and reduces repetitive manual effort.	Can produce incorrect or unsupported answers if not validated.
Helps users interact with complex systems in natural language.	Needs careful design of prompts, data access, permissions, and monitoring.
Can support training, support, coding, analysis, and document workflows.	May have cost, latency, privacy, and governance concerns in production.

6. Best Practices

- Start with a simple prototype before building a complex system.
- Use clear prompts and structured output formats.
- Add validation and human review for important decisions.
- Measure quality with realistic test cases, not only demo examples.
- Monitor cost, latency, accuracy, safety, and user feedback.
- Document assumptions, limitations, and escalation paths.

7. Key Takeaways

Retrieval Augmented Generation (RAG) is an important concept in modern LLM and GenAI systems. The simple idea is easy to understand, but production implementation requires thoughtful architecture, data quality, evaluation, safety, and monitoring.

Quick revision

- Simple meaning: RAG means connecting an LLM with external documents or databases. Instead of answering only from memory, the system first retrieves relevant information and then generates a grounded answer.
- Architecture: input, processing, tools/data, validation, output.
- Business value: saves time, improves knowledge access, and supports automation.
- Risk control: use grounding, evaluation, guardrails, monitoring, and human approval.